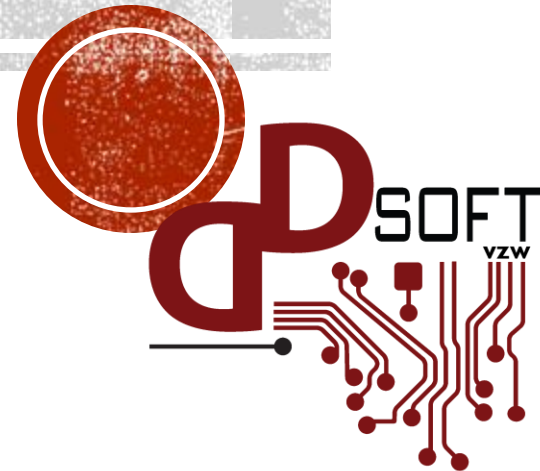


MAKING ACCESSIBLE APPS IN XAMARIN

2018 – Dennie Declercq



WHO AM I?



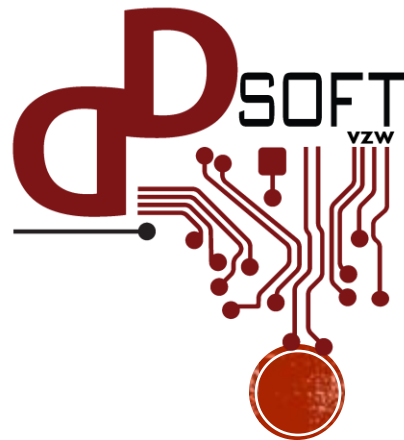
Dennie Declercq

Founder of DDSoft nonprofit.

Volunteer developer at Ithaka.

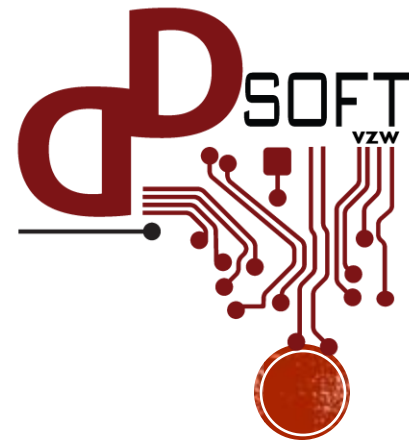
Honored to speak here!

Inventor of the hashtag #DreamingIsBelieving



CONTENT

- Eye openers
- The basics of Xamarin.Forms
- The basics of accessibility
- Inclusive layout
- Demo Time
- Recap
- Q&A

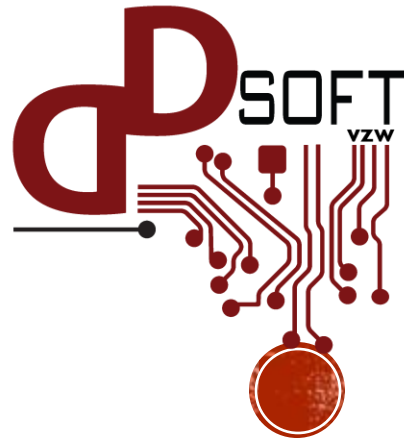


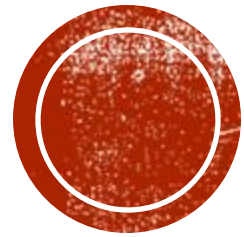
TECHNOLOGY WE USE

For practical reasons, we build only for **Android** and **Windows 10** at this time!



Xamarin





EYE OPENERS

Let's start with a few "eye openers".



CAN YOU REALIZE ?

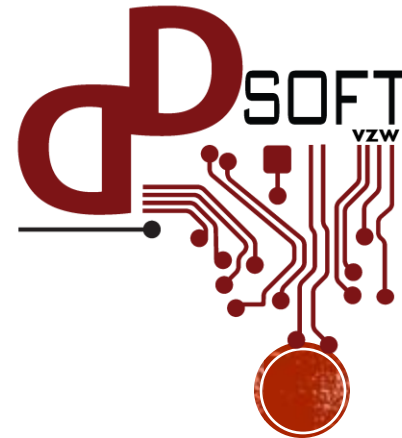
*That the things that we're all do...
..Are not common "life" for many people?*

Learning disability special

"Research has shown that with brain training exercises, children are able to learn and read faster, more easily and with greater comprehension," Du Plessis added.

Edublox, a South African company that helps children and adults with learning disabilities through specially developed programmes, have introduced a bursary programme for pupils with learning disabilities in less

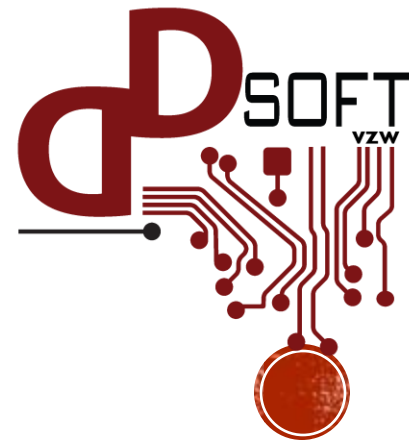
fortunate communities. "Not being able to learn is debilitating and learners from these areas have a lot to contend with as it is," said Zainu Allie, Edublox franchise owner and facilitator in Cape Town. "We have been working closely with 16 learners from schools in the Hout Bay area since April. With the safe, fun and challenging environment provided, at our Centre in Claremont, we have seen

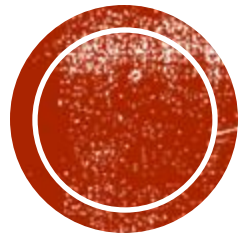


WHAT'S THIS?



An
eating
schedule





THE BASICS OF XAMARIN.FORMS



THE NEW PROJECT LAYOUT

New Cross Platform App - EmoBoard

Select a template:


Blank App


Master Detail

A cross-platform template for building cross-platform, native mobile apps for iOS, Android, and Universal Windows Platform.

Use the native user interface approach to create UIs for each platform, or use Xamarin.Forms to create cross-platform, native UIs in XAML.

UI Technology

☒ Xamarin.Forms

☐ Native

Code Sharing Strategy ?

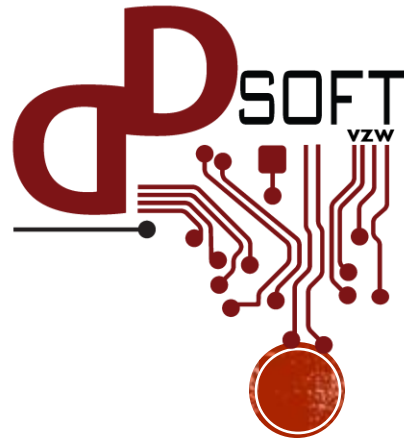
☐ Shared Project

☒ Portable Class Library (PCL)

Microsoft Azure

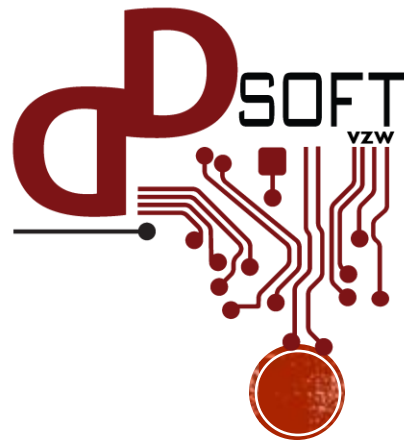
☐ Host in the cloud

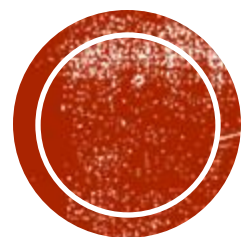
OK Cancel



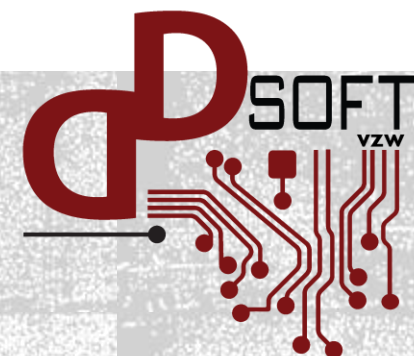
TODO

- Introduction of xamarin





THE BASICS OF ACCESSIBILITY



DIFFERENT KINDS OF DISABILITIES

Physical

- Can't move like the 'normal way'.

Mental / Intellectual

- Can't think / speak / read / behave (communicate) like the 'normal way'.

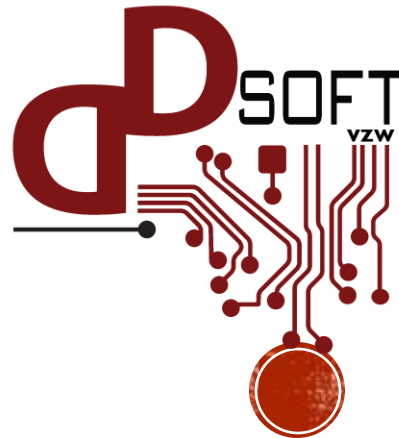
Visual

- They can see like 'the normal way'.

Audio

- They can't hear like 'the normal way'.

Other (specific and combinations)



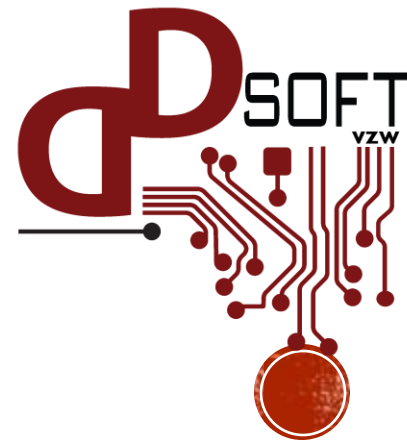
VISION IMPAIRMENT

Decreased ability to:

- Read
- Navigate
- Navigate the web
- Operate a smartphone

Technology that helps

- Large fonts, Text To Speech
- GPS with TTS
- Keyboard Access
- Browser with TTS
- Narrator
- Magnifier
- High contrast
- Voice commands
- Screenreaders



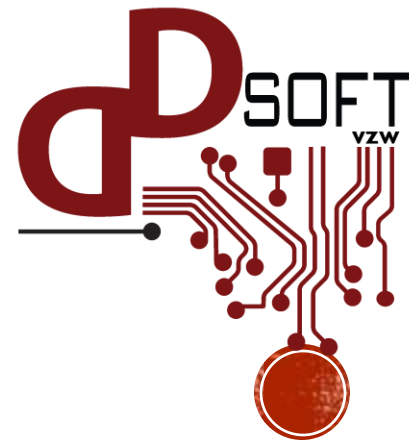
AUDITIVE IMPAIRMENT

Difficulties

- Hear notifications
- Understand video's

Technology that helps

- Notifications with buzz
- Notifications staying active
- Subtitles in movies



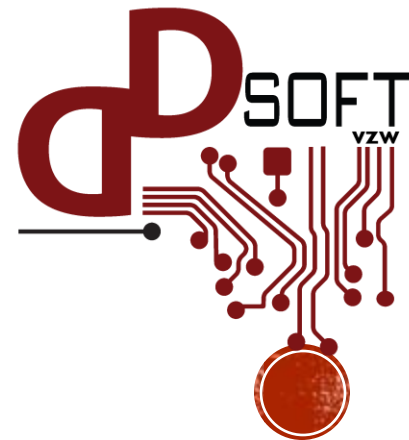
MOTORIC IMPAIRMENT

Difficulties

- Control a mobile device

Technology that helps

- Eye camera
- Touch screen



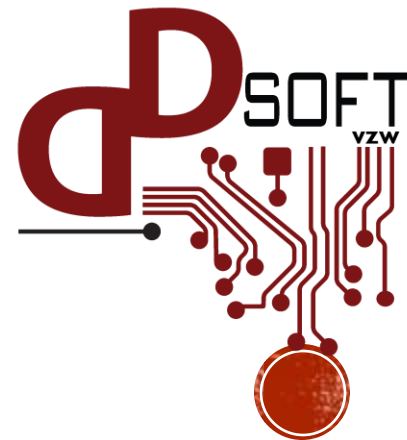
INTELLECTUAL DISABILITY

Difficulties

- Understand written text
- Understand lay-out of app / software

Technology that helps

- Simplified layout
- Structure
- Text-To-Speech
- Pictures and pictographs



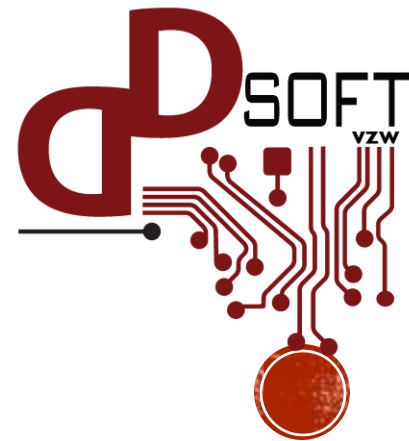
AUTISM

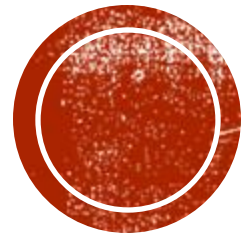
Difficulties

- Understand layout of application or app
- Distraction

Technology that helps

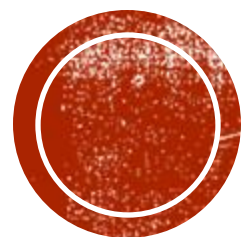
- Simplified layout
- Engagement triggers
- Structure
- Pictures and pictographs



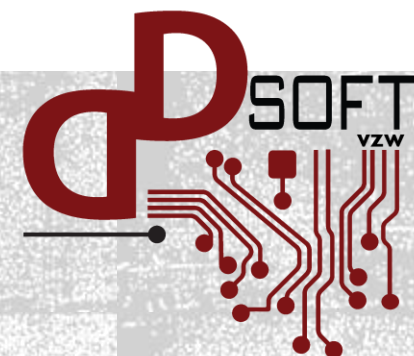


FOCUS OF TODAY: INTELLECTUAL DISABILITIES (AND AUTISM)

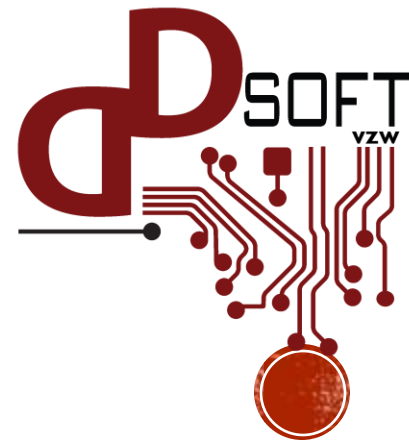




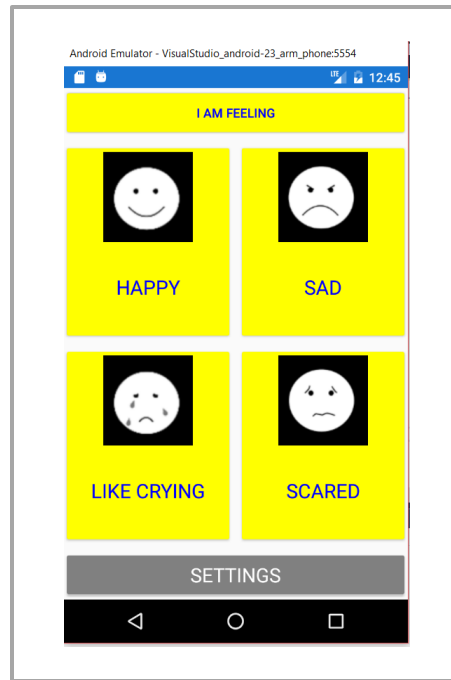
**THERE ARE CA 2-3% PEOPLE
WITH AN INTELLECTUAL
DISABILITY IN THE WORLD.**



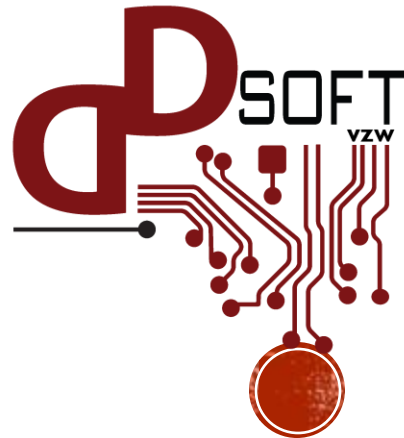
LET'S LEARN TO MAKE APPS FOR PEOPLE WITH A DISABILITY IN XAMARIN.FORMS!



DEMO 'EMOBOARD'!

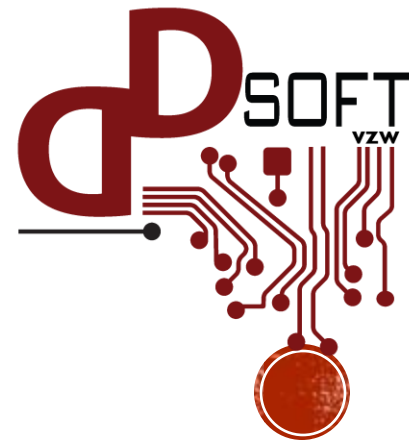


“Many people with a disability have difficulties to talk about their mood.”



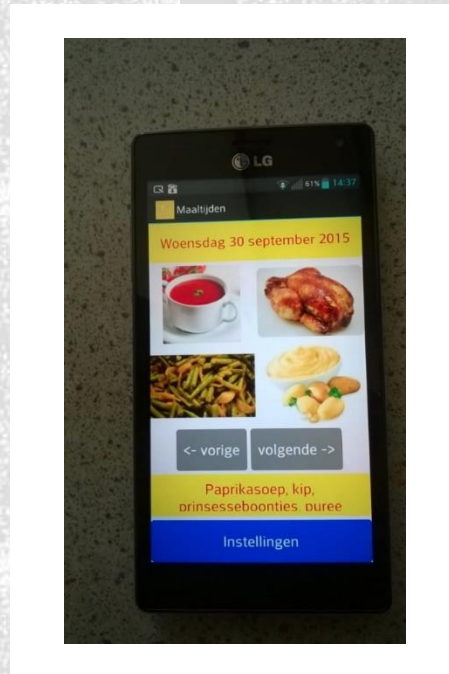
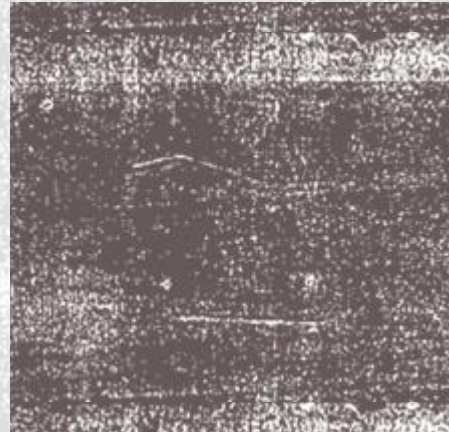
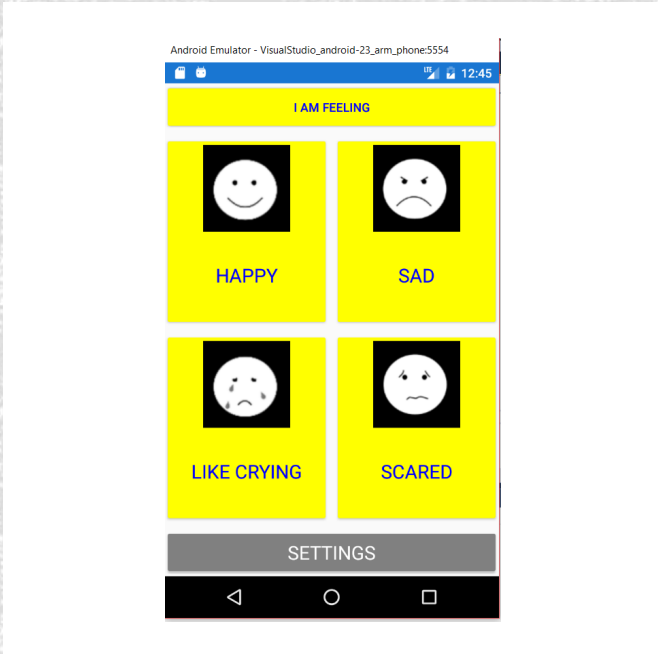
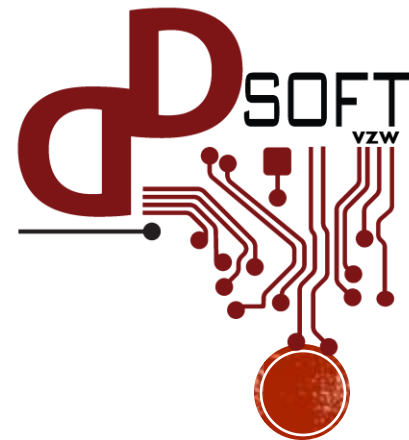
BASIC RULES

1. Simplify the layout;
2. Use pictographs or pictures;
3. Use Text-To-Speech;
4. (optional) An inclusive approach.



SIMPLIFY THE LAYOUT

- Light background color
- Accent colors
- **Color by function**
- Structure
- Easy screens
- Be consistent!

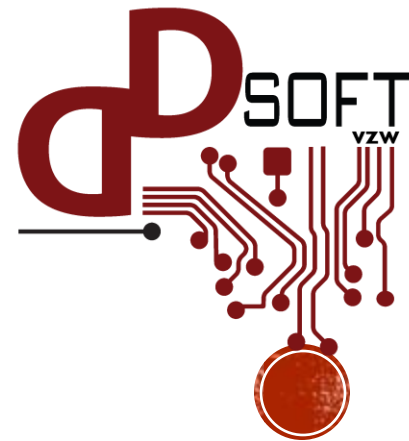


COLOR BY FUNCTION

I can speak !

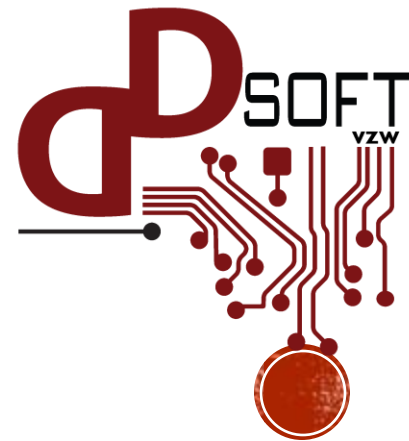
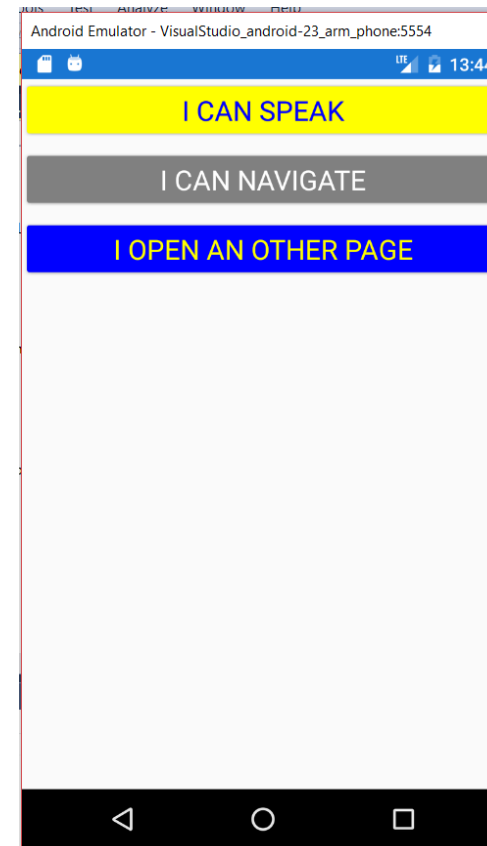
I can navigate !

I open an other page!



COLOR BY FUNTION (IN CODE)

```
<StackLayout>
  <Button x:Name="SpeakButton"
    BackgroundColor="Yellow"
    TextColor="Blue"
    FontSize="Large"
    Text="I can speak"
    Clicked="SpeakButton_Clicked"/>
  <Button x:Name="NavigateButton"
    BackgroundColor="Gray"
    TextColor="White"
    FontSize="Large"
    Text="I can navigate"
    Clicked = "NavigateButton_Clicked"/>
  <Button x:Name="OtherPageButton"
    BackgroundColor="Blue"
    TextColor="Yellow"
    FontSize="Large"
    Text="I open an other page"
    Clicked = "OtherPageButton_Clicked"/>
</StackLayout>
```

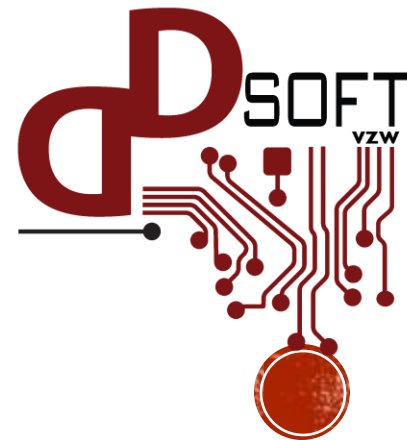


USE PICTOGRAPHS OR PICTURES

- What means the following?

- **Kip**

- The answer is:

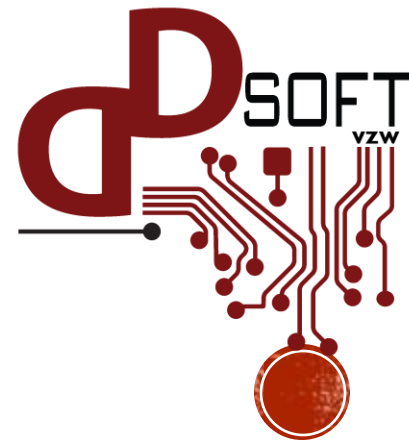
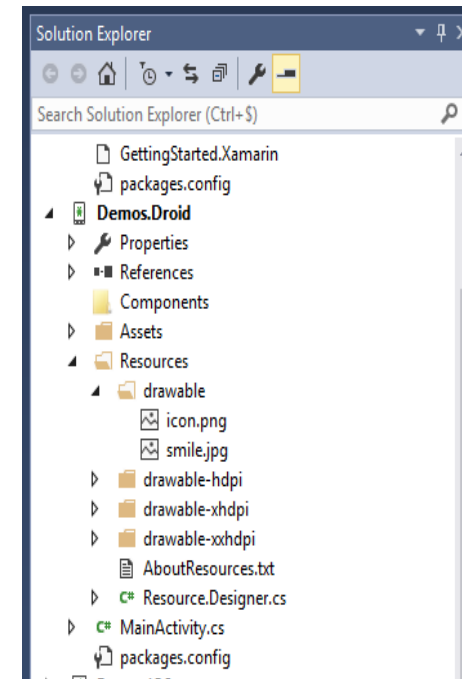


USE PICTOGRAPHS OR PICTURES (IN CODE)

```
ContentLayout="Top,0"
```

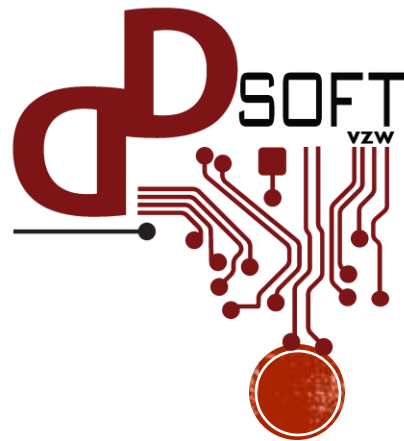
```
Image="Happy.png"
```

“ContentLayout” is needed to be sure that the image is on top and the text is below.
It’s a good practice to show also the text!



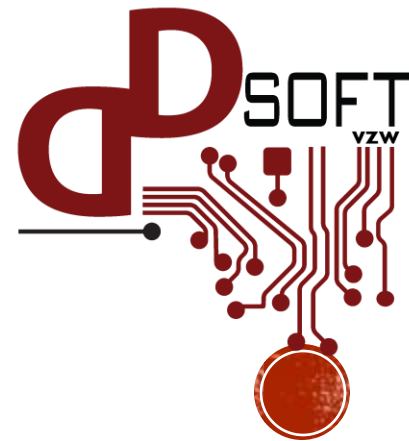
TEXT-TO-SPEECH

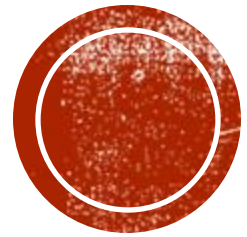
- With Xamarin.Forms.Essentials!



THE DIFFERENCE BETWEEN NARRATOR AND TEXT TO SPEECH

- **Narrator** reads **sequential** every control
- **Text To Speech** reads the control where **Tapped On**
- **Narrator** is good for people with **vision loss**
- **Narrator** is **confusing** for people with **intellectual disabilities**



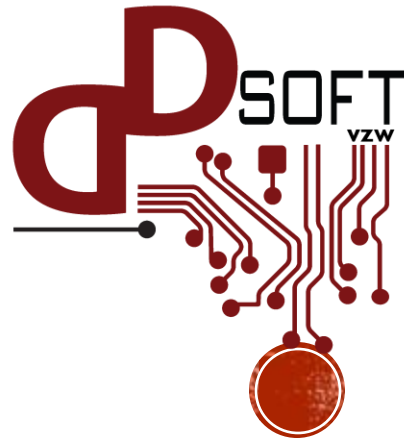


INCLUSIVE LAYOUT

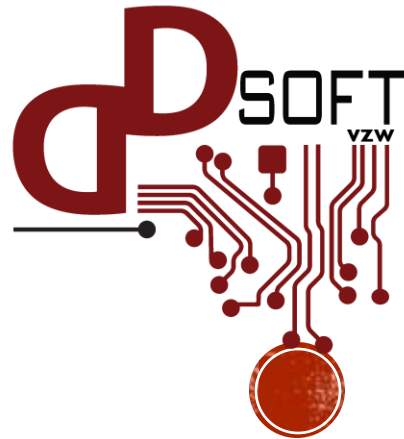
For people who doesn't need all the functions



ADDING THE SETTINGS PAGE



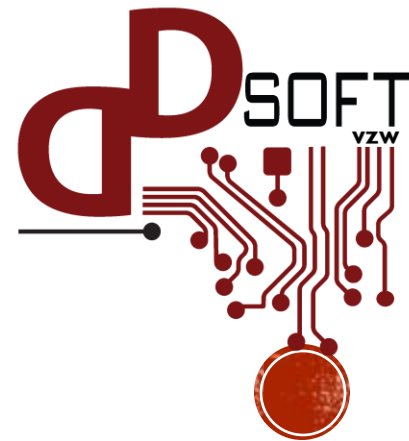
MODYFING APPCS



WE ADD THE SETTINGS.CS CLASS

- The **settings.cs** class has the following code:

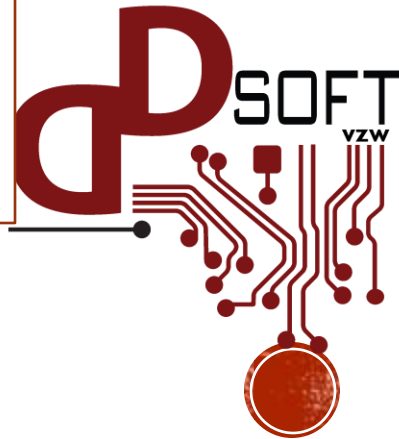
```
public static class Settings
{
    public static bool EnableTTS = true;
}
```



THE CODE BEHIND OF SETTINGSPAGE.CS

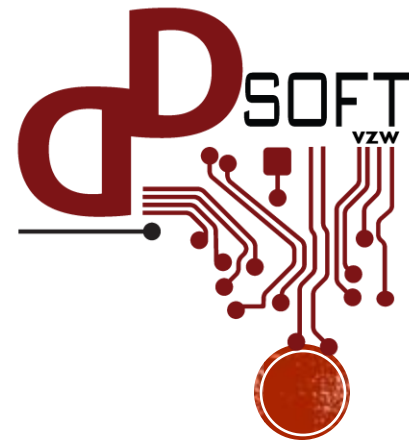
```
if(TTSToggle.IsToggled == true)
{
    Settings.EnableTTS = true;
} else
{
    Settings.EnableTTS = false;
}

await Navigation.PopToRootAsync();
```



MODIFYING OUR EXISTING CODE.

```
private void HappyButton_Clicked(object sender, EventArgs e)
{
    if(Settings.EnableTTS == true)
    {
        CrossTextToSpeech.Current.Speak(HappyButton.Text.ToString());
    }
}
```



SMOOTHING DOWN THE ROUGH EDGES (SETTINGSPAGE.XAML.CS)

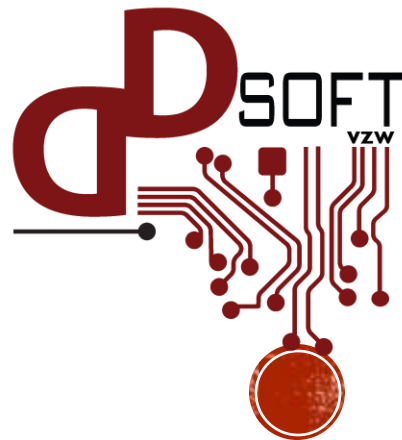
- Since we have a **Navigation Page**, we must set a **Title**.
- We must load the state of the **Toggle button**, otherwise it's always true when coming back on the page.

```
this.Title = "Settings";  
if(Settings.EnableTTS == true)  
{  
    TTSToggle.IsToggled = true;  
} else  
{  
    TTSToggle.IsToggled = false;  
}
```



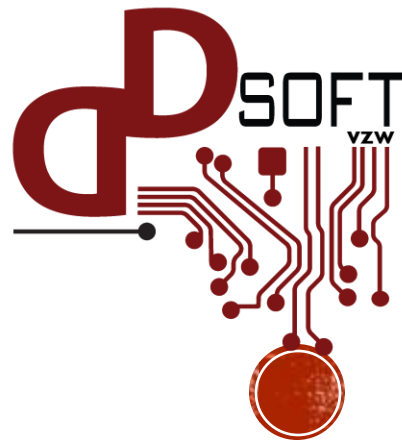
SMOOTHING THE ROUGH EDGES (MAINPAGE.XAML.CS)

- Also here, we have to make the page have a **title**.



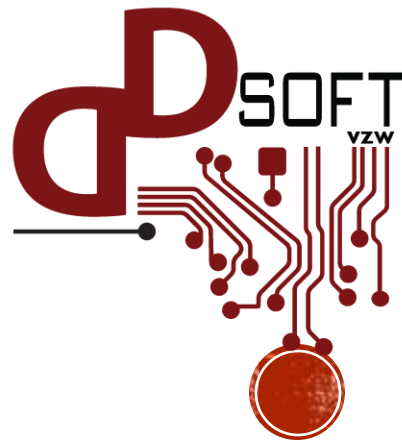
DEMO: ENABLE CHOICES (INCLUSIVE APPROACH)

- Final demo



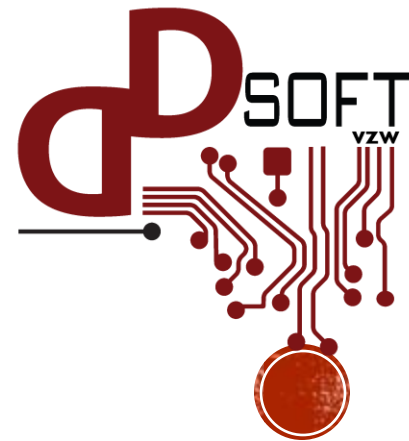
THE LAST THING THAT ISN'T IMPLEMENTED

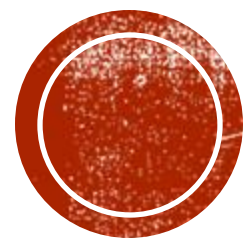
- Settings must be saved in a database like **SQLite** or a **Settings Plugin**.



RECAP

1. Simplify the layout with color by function
2. Use images and pictographs
3. Use TTS
4. Add an inclusive approach

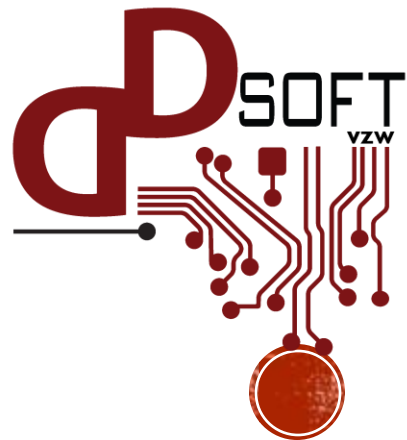




NOW IT'S UP TO YOU!



Q&A



THANKS!

- I am Dennie Declercq.
- I live in Belgium.
- My e-mail adress is: dennie@ddsoft.be.
- You can call me: +32478294525.
- You can help me spread the word on Twitter:
@DennieDeclercq.

